

Network 1 : Logistics 8000 hosts

$$2^{13} = 8192 - 2 = 8190 > 8000$$

100.22.0.0 /19 255.255.224.0 network

100.22.0.1/19 100.22.31.254/19 hosts

100.22.31.255/19 broadcast

100.22.32.0/19 next network

Network 2 : Management 4000 hosts

$$2^{12} = 4096 - 2 = 4094 > 4000$$

100.22.32.0 /20 255.255.240.0 network

100.22.32.1/20 100.22.47.254/20 hosts

100.22.47.255/20 broadcast

100.22.48.0/20 next network

Network 3 : human resources 4000 hosts

$$2^{12} = 4096 - 2 = 4094 > 4000$$

100.22.48.0/20 255.255.240.0 network

100.22.48.1/20 100.22.63.254/20 hosts

100.22.63.255/20 broadcast

100.22.64.0/20 next network

Network 4 : Customer Care 3000 hosts

$$2^{12} = 4096 - 2 = 4094 > 3000$$

100.22.64.0 /20 255.255.240.0 network

100.22.64.1/20 100.22.79.254/20 hosts

100.22.79.255/20 broadcast

100.22.80.0/20 next network

Network 5 : Guest area 2000 hosts

$$2^{11} = 2048 - 2 = 2046 > 2000$$

100.22.80.0 /21 255.255.248.0 network

100.22.80.1/21 100.22.87.254/21 hosts

100.22.87.255/21 broadcast

100.22.88.0/21 next network

Network 6 : Research 1800 hosts

$$2^{11} = 2048 - 2 = 2046 > 1800$$

100.22.88.0 /21 255.255.248.0 network

100.22.88.1/21 100.22.95.254/21 hosts

100.22.95.255/21 broadcast

100.22.96.0/21 next network

Network 7 : Marketing 1600 hosts

$$2^{11} = 2048 - 2 = 2046 > 1600$$

100.22.96.0 /21 255.255.248.0 network

100.22.96.1/21 100.22.103.254/21 hosts

100.22.103.255/21 broadcast

100.22.104.0/21 next network

Network 8 : Accounting 1200 hosts

$$2^{11} = 2048 - 2 = 2046 > 1200$$

100.22.104.0 /21 255.255.248.0 network

100.22.104.1/21 100.22.111.254/21 hosts

100.22.111.255/21 broadcast

100.22.112.0/21 next network

Network 9 : Accounting 850 hosts

$$2^{10} = 1024 - 2 = 1022 > 850$$

100.22.112.0 /22 255.255.252.0 network

100.22.112.1/22 100.22.115.254/22 hosts

100.22.115.255/22 broadcast

100.22.116.0/22 next network

Network 10 : ICT 120 hosts

$$2^7 = 128 - 2 = 126 > 120$$

100.22.116.0 /25 255.255.255.128 network

100.22.116.1/25 100.22.116.126/25 hosts

100.22.116.127/25 broadcast

100.22.116.128/25 next network

Network 11 : Finance 60 hosts

$$2^6 = 64 - 2 = 62 > 60$$

100.22.116.128 /26 255.255.255.192 network

100.22.116.129/26 100.22.116.190/26 hosts

100.22.116.191/26 broadcast

100.22.116.192/26 next network

Network 12 : server room 3 hosts

$$2^3 = 8 - 2 = 6 > 3$$

100.22.116.192 /29 255.255.255.248 network

100.22.116.193/29 100.22.116.198/29 hosts

100.22.116.199/29 broadcast

100.22.116.200/29 next network

